Sahaj Bhadja

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Summary

Product-focused engineer experienced in **technical issue investigation, systems analysis, simulation, verification, validation, testing, risk mitigation, and documentation**, using data-driven decisions to improve reliability, performance, and quality.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Investigated performance and reliability issues using state tracking, validation, reconciliation, and structured error handling, increasing concurrent throughput 3x while reliably sustaining 99%+ reliability across 5,000+ monthly jobs.
- Evaluated system requirements and data interactions for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through schema analysis, indexing, caching, and measured performance optimization.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Diagnosed OCR, model, ingestion, and API failures across 10,000+ daily requests, implementing validated fallback paths, diagnostics, and automated tests while reliably reducing end-to-end feedback turnaround time 60%.
- Developed cost-aware validation workflows processing 1,000+ files per batch, tracking model usage, token consumption, and quality outcomes while maintaining stable performance under concurrent production workloads at scale.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Modeled and simulated system interactions across 100,000+ trajectory records, improving long-horizon planning 18% through performance analysis, reproducible testing, edge-case evaluation, debugging, and iterative technical optimization at scale.
- Evaluated closed-loop behavior across ego, road, and nearby-agent signals, reliably improving stability 15% while documenting technical tradeoffs, experimental results, interface interactions, and data-driven recommendations for senior researchers.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Investigated integration defects and implemented permanent API improvements through filtering, pagination, caching, and validation, improving response times 30%, reducing payload sizes 25%, and lowering technical debt 20%.
- Planned and executed 20+ unit and integration tests across Windows, macOS, and Linux, documenting failures and coordinating improvements with Microsoft and GitHub stakeholders during Agile reviews and deployments.

Cyber Security and Data Mining: Detecting Malicious URL **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*
- Designed 21 statistical indicators across 650,000 URLs, evaluating IP presence, suspicious keywords, and structural anomalies to identify potential failure patterns, security risks, phishing, and malware accurately at scale.
- Benchmarked Random Forest, recurrent, and convolutional models, achieving 96%+ detection accuracy while reliably comparing performance tradeoffs, validating results, and documenting reproducible findings from malicious and benign datasets.

Skills

Engineering: Systems thinking, requirements analysis, interface analysis, risk assessment, documentation, collaboration
Quality: Root-cause investigation, failure isolation, verification, validation, unit testing, integration testing, reliability
Analytics: Python, SQL, statistical analysis, feature engineering, simulation, modeling, performance analysis, data visualization
Tools: Git, CI/CD, Jenkins, PyTest, Mockito, Pandas, NumPy, PyTorch, Matplotlib, SQL Server

Projects

CPU-scheduling Simulator Live | *System Performance Analysis*
- Implemented and verified six C++ scheduling algorithms with edge-case handling, automated performance metrics, and Gantt visualization, testing 1,000+ process combinations while reliably reducing average turnaround time 25%.

CaptionCraft - Image Caption Generator Live | *Design Evaluation and Validation*
- Benchmarked BLIP, GIT, ViT-GPT-2, and custom architectures across 10,000 images and 100,000 captions, validating design alternatives and improving measured accuracy from 81% to 96% through comparative analysis.

Comprehensive Analysis of Blockchain for User Authentication | *Risk and Systems Research*
- Evaluated decentralized authentication across three access scenarios and four permission tiers, identifying interface risks and designing tamper-resistant workflows that reduced repeated key-verification steps by 50% through structured comparative analysis.